

Class XII Chemistry

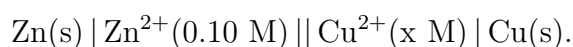
JEE Advanced-style Practice Examination

Time Allowed: 180 minutes**Maximum Marks:** 25**Name:** _____**Roll No.:** _____

General Instructions

- All questions are compulsory. Each question carries 5 marks.
- Show the essential steps, balanced equations, and units wherever applicable.
- Use $E^\circ_{\text{cell}} = 1.10 \text{ V}$ for the Zn/Cu cell and $2.303RT/F = 0.0591 \text{ V}$ at 298 K.

1. Consider the galvanic cell



At 298 K, the observed cell potential is 1.07 V. Calculate x , the concentration of Cu^{2+} . [5 marks]

2. A first-order reaction is 75% complete in 20 minutes. Calculate:

1. the rate constant k , and
2. the time required for 93.75% completion.

[5 marks]

3. An organic compound A has molecular formula $\text{C}_3\text{H}_6\text{O}$. It gives a positive 2,4-DNP test but does not reduce Tollens' reagent. On reduction with NaBH_4 , it gives compound B . Dehydration of B with concentrated H_2SO_4 on heating gives propene.

1. Identify A and B .
2. Write the reactions for $A \rightarrow B$ and $B \rightarrow$ propene.

[5 marks]

4. Compare the complexes $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$. For each complex, state:

1. the d -electron configuration of the metal ion,
2. the number of unpaired electrons,
3. the hybridisation, and
4. whether it is paramagnetic or diamagnetic.

[5 marks]

5. 100 mL of 0.020 M BaCl_2 is mixed with 100 mL of 0.010 M Na_2SO_4 . Given

$$K_{\text{sp}}(\text{BaSO}_4) = 1.1 \times 10^{-10},$$

determine whether a precipitate forms. Calculate the equilibrium concentration of SO_4^{2-} and the mass of BaSO_4 precipitated. ($M(\text{BaSO}_4) = 233 \text{ g mol}^{-1}$) [5 marks]